ViT-PCOS: Vision Transformer pour la détection automatisée du Syndrome des Ovaires Polykystiques

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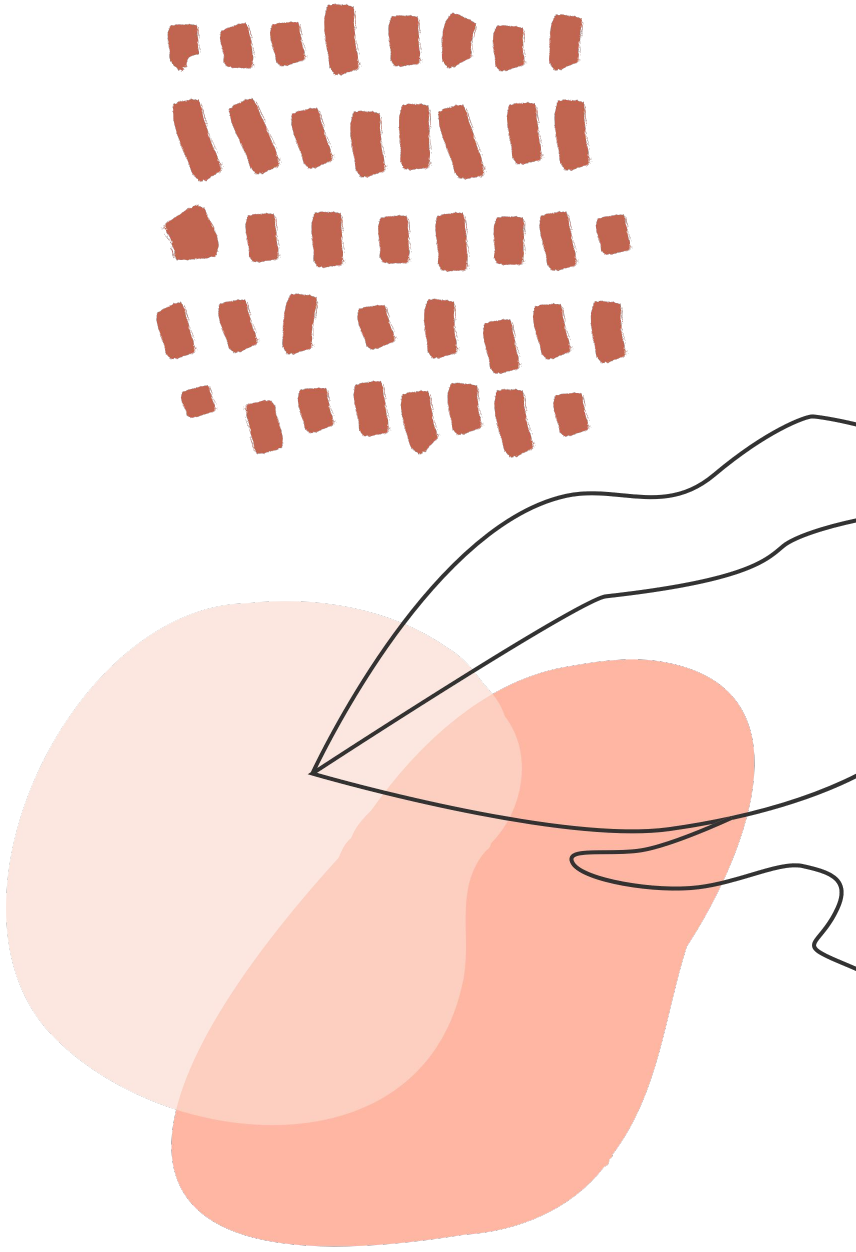
Introduction

Saviez-vous que?

- 8 à 13 % des femmes en âge reproductif ont le SOPK.
- 70% pas diagnostiquer
- Lié au diabète de type 2 et au diabète gestationnel.

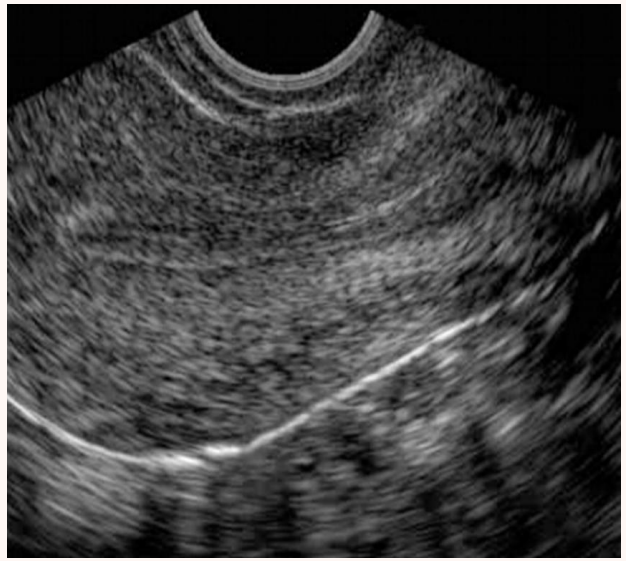
Problème :

- Faciliter le diagnostic et rendre plus rapide
- Communautés mal desservies.



Données

SOPK



Non-SOPK

Info

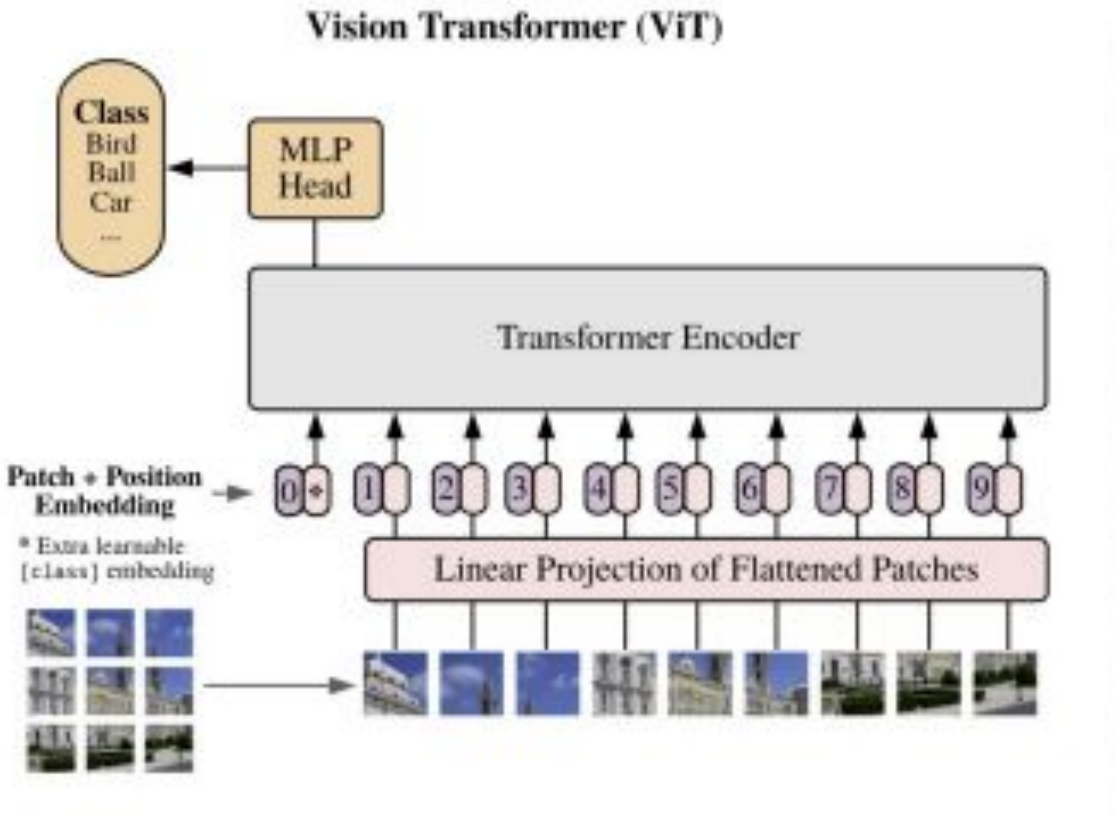
Directory	Infected Images	Non-Infected Images
Train	781	1143
Test	787	1145

Total Test Images: 1932
Total Train Images: 1924

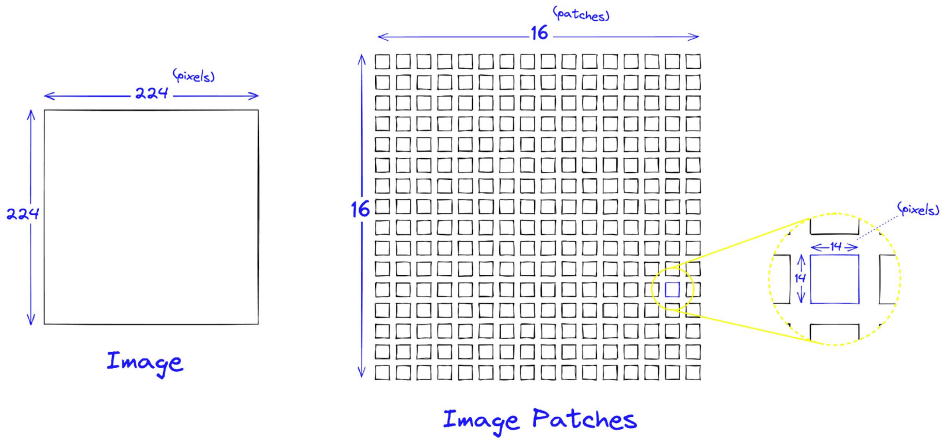
Résumé de la théorie

Préparation du Dataset	ViT-L-16	Transformer Encoder	Classification Head	Entrainement
infecter VS non infecter	Image diviser en patches de 16X16 pixel	MSA MLP	6 couches ReLu activation	Vision Transformer (ViT-L-16) pre-trained on ImageNet-21k dataset Cross-Entropy Loss function

ViT-L-16



1. Linear patches 16X16 pixel
2. Vecteur dimension d
3. Positional encoding pour preserver layout spatial



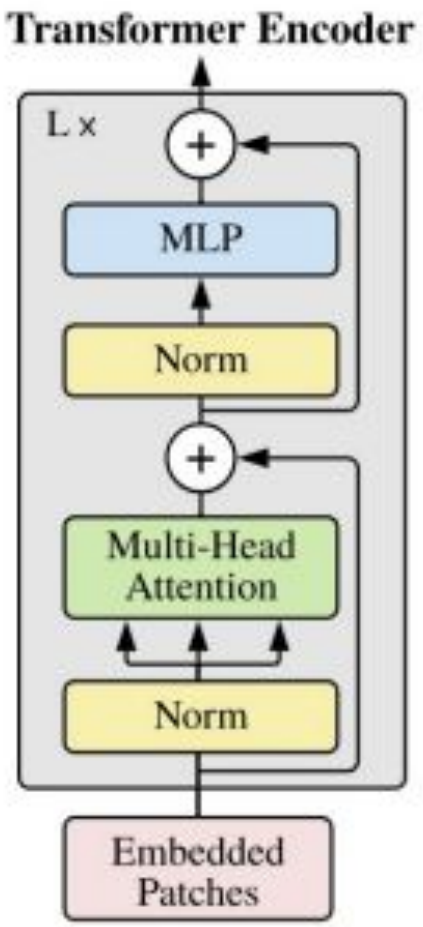
Transformer Encoder

MSA: multi-head self-attention block

- va déterminer les relations entre les patches (linéaire, self-attention, concaténation, linéaire encore)

MLP: feed-forward dense block

- GeLu



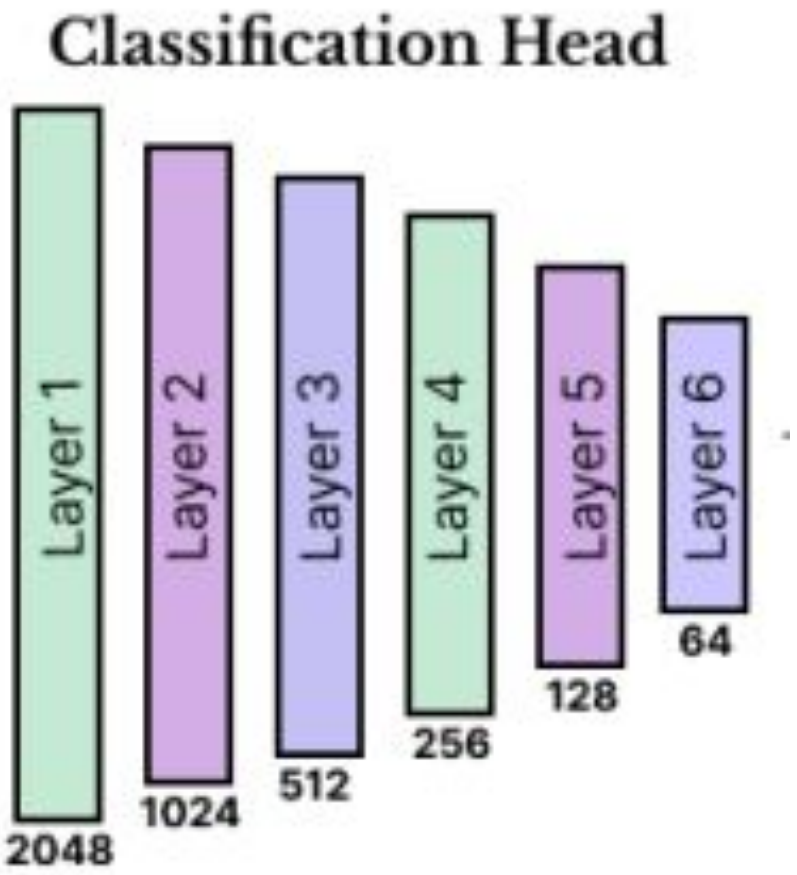
L : large nombre de parametres

```
# configuration du modele avec transformer encoder
config = ViTConfig(
    image_size=IMG_SIZE,
    num_channels=3,
    patch_size=16,
    num_classes=NUM_CLASSES,
    hidden_size=768,
    num_hidden_layers=12,
    num_attention_heads=12,
    intermediate_size=3072,
    hidden_dropout_prob=0.1,
    attention_probs_dropout_prob=0.1,
)
```

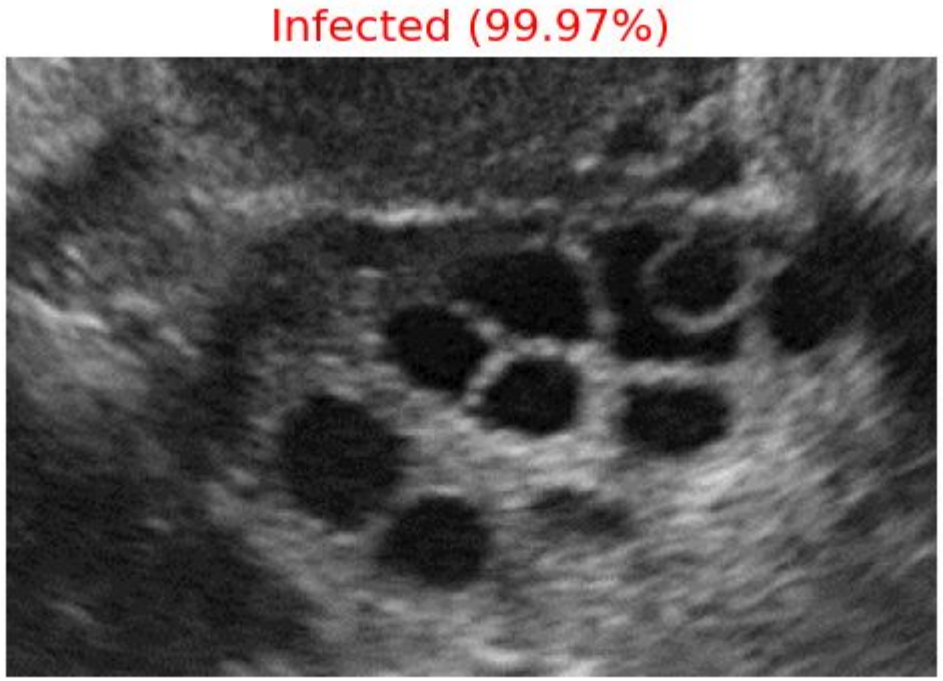
Classification Head

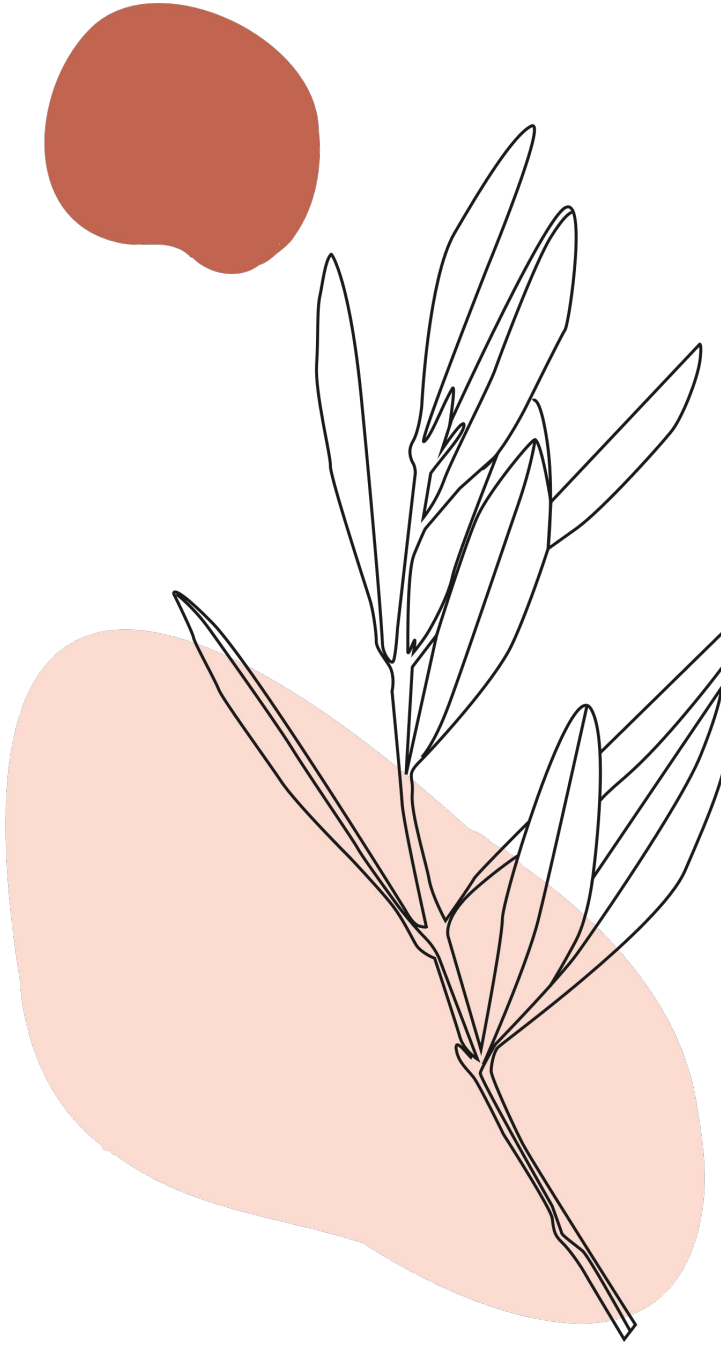
- Caractéristique complexe
- Bottleneck (Goulot d' étranglement)

```python
class CustomViTForImageClassification(nn.Module):
    def __init__(self, config):
        super(CustomViTForImageClassification, self).__init__()
        self.vit = ViTModel(config)
        self.classifier = nn.Sequential(
            nn.Linear(config.hidden_size, 2048),
            nn.ReLU(),
            nn.Linear(2048, 1024),
            nn.ReLU(),
            nn.Linear(1024, 512),
            nn.ReLU(),
            nn.Linear(512, 256),
            nn.ReLU(),
            nn.Linear(256, 64),
            nn.ReLU(),
            nn.Linear(64, config.num_classes)
        )
```



Prédictions





Résultats

- 100% précision sur le set de test
- 97% précision avec CNN

TABLE II
SUMMARY OF STUDIES ON OVARIAN IMAGING (NA DENOTES NOT AVAILABLE)

Paper	Dataset	No. of Images	Model	Accuracy
kiruthika et al [13]	Data from the Department of Obstetrics & Gynaecology, Chettinad Academy of Research & Education after proper clearance from the Ethical Committee of the institute.	635	Machine Learning	96%
nabilah et al [19]	Data collected through surveys and provides guidance to Obstetrics and Gynecology (Obgyn) specialists	46	Watershed Segmentation and Contour Analysis	97.8%
Madhumitha et al [17]	Public dataset from Radiologykey Platform	171	Hybrid method using SVM, KNN, Logistic Regression	98%
Rachana et al [21]	NA	50	KNN Classifier	97%
Gopalakrishnan et al [10]	NA	NA	Integrated Transfer Learning CNN	98.9%
Begam et al [3]	NA	80	CNN	94%
Hosian et al [11]	PCOS detection using ultrasound images - Kaggle dataset	3856	CNN	98.12%
Chitra et al [4]	NA	1600	Resnet50, VGG16, Hybrid	93%
Dewi et al [7]	Data from Klinik Bersalin Permata Bunda Syariah, Cirebon	NA	Neural Network	80.84%
This work	PCOS detection using ultrasound images - Kaggle dataset	**3846**	Vision Transformer	**100 %**

Conclusion

Montre le potentiel pour un déploiement réel dans les systèmes de support à la décision clinique.
Mieux que CNN

Tester avec un dataset plus grand

Compter le nombre de kystes

Détecter le stade du SOPK (sévère ou non)

Questions & Reponses

References

- https://huggingface.co/docs/transformers/en/model_doc/vit
- https://discuss.huggingface.co/t/how-do-i-change-the-classification-head-of-a-model/4720/12
- https://ieeexplore-ieee-org.proxy.bibliotheques.uqam.ca/document/10866446
- https://github.com/AarohiSingla/Image-Classification-Using-Vision-transformer/blob/main/image_classifier_from_scratch.ipynb